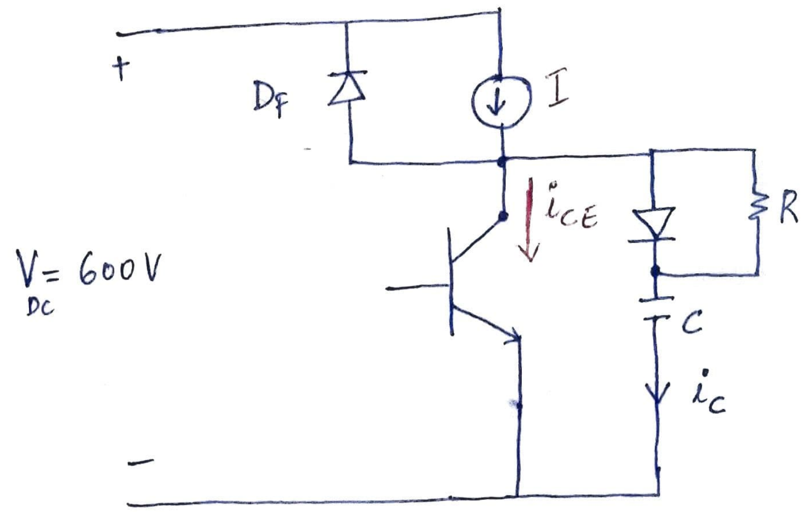
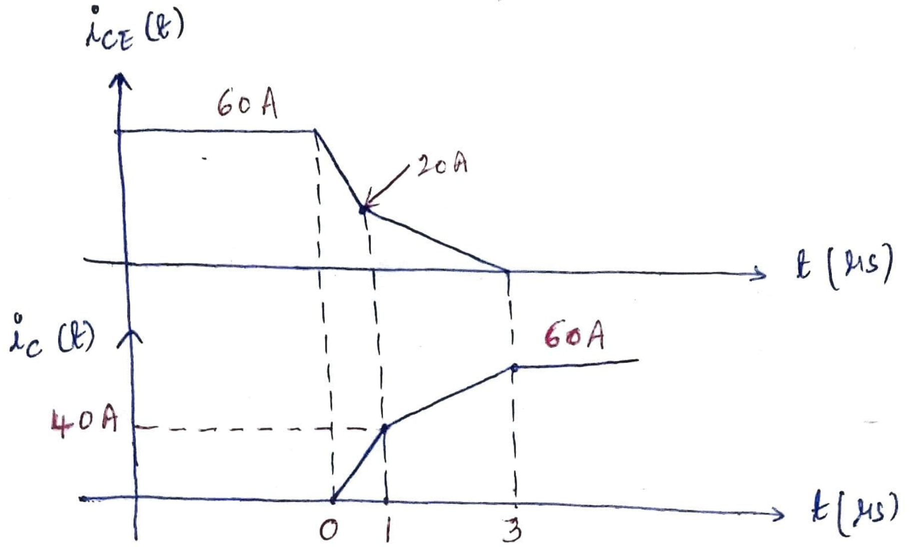
ASSIGNMENT - 2

(1) Given, $V_{DC} = 600\ V$; $I = 60A$, $C = 0.5\mu F$



(A)



(B) $0 \le t \le 1\mu s$:-

$$i_{CE}(t) = 60 - 40\times 10^6 t \ . \ A$$

$$i_c(t) = I - i_{CE}(t) = 60 - 60 + 40\times 10^6 t$$

$$i_c(t) = 40\times 10^6 t \quad A.$$

$$\therefore \; v_c(t) = \frac{1}{C}\int_0^t i_c(t)\,dt$$

$$= \frac{1}{C}\int_0^t 40\times10^6\, t\, dt$$

$$= \frac{40\times10^6}{C}\cdot\frac{t^2}{2} = \frac{40\times10^6}{2\times0.5\times10^{-6}}\, t^2$$

$$\boxed{v_c(t) = 40\times10^{12}\, t^2 \;\; V}$$

@ $t = 1\,\mu s$

$v_c(t) = 40\,V$.

$1\,\mu s \le t \le 3\,\mu s$:-

$$i_{CE}(t) = 30 - 10\times10^6\, t \;\; A$$

$$i_c(t) = I - i_{CE}(t) = 60 - 30 + 10\times10^6\, t$$

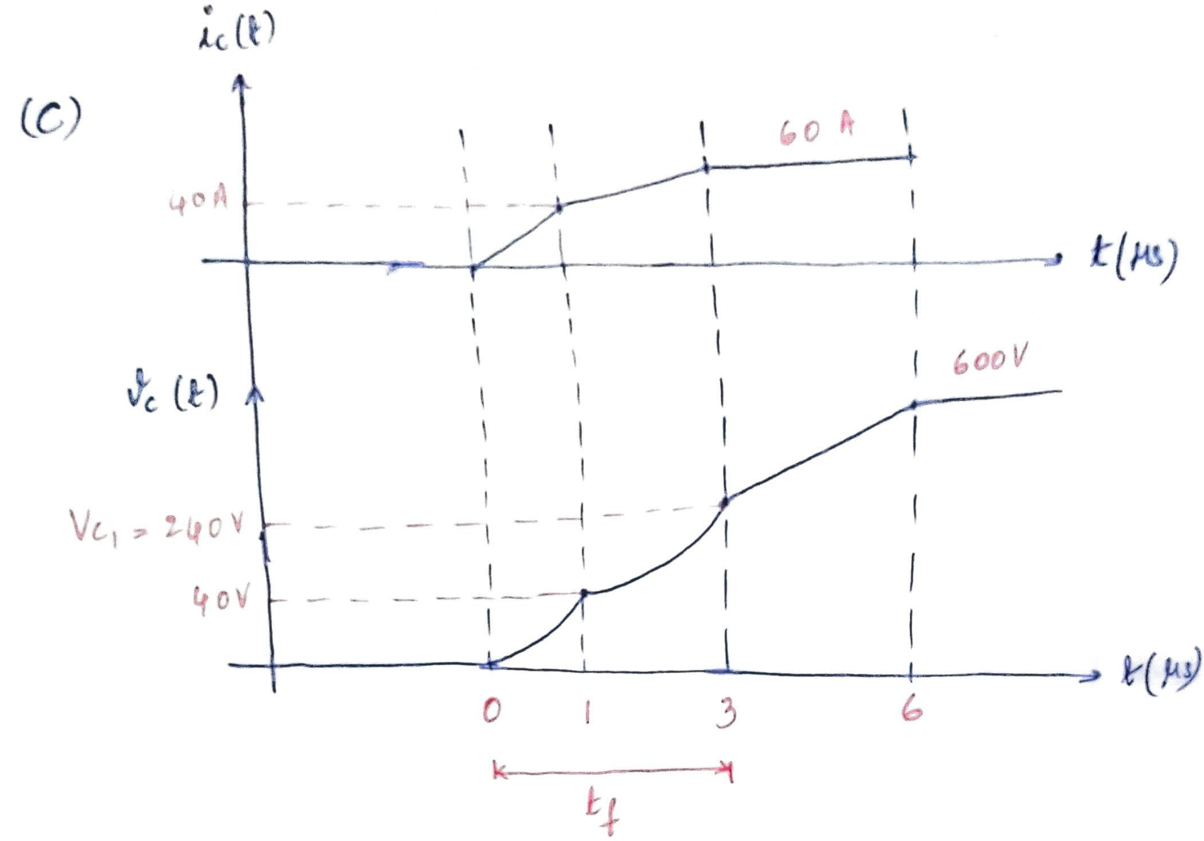
$$i_c(t) = 30 + 10\times10^6\, t \;\; A$$

$$\therefore \; v_c(t) = \frac{1}{C}\int_{1\times10^{-6}}^{t} i_c(t)\,dt + v_{c,\text{initial}}$$

$$= \frac{1}{C}\int_{10^{-6}}^{t} 30 + (10\times10^6\, t)\,dt + 40$$

$$\boxed{\therefore \; v_c(t) = 10\times10^{12}\, t^2 + 60\times10^6\, t - 30 \;\; V}$$

(C)



(D) $3\mu s \le t \le t_1$:-

$$v_c(t) = V_{C_1} + \frac{V_{DC} - V_{C_1}}{(t_1 - 3)\times 10^{6}} (t - 3\times 10^{6})$$

$$= V_{C_1} + \frac{I}{C}(t - 3\times 10^{6}) \quad \left[\because \frac{\Delta V}{\Delta t} = \frac{I}{C}\right]$$

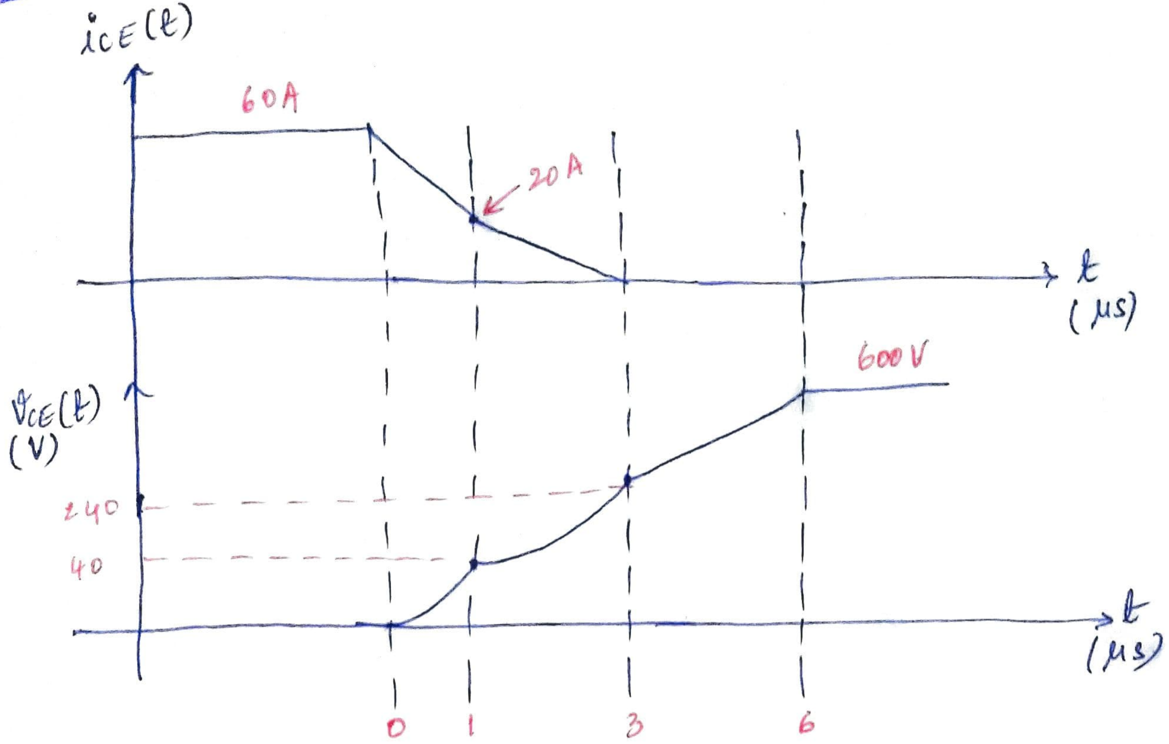
At $t = t_1$; $V_c(t) = 600\ V$

$$\therefore\ 600 = 240 + \frac{60}{0.5\times 10^{-6}} (t_1 - 3\times 10^{6})$$

$$\boxed{\therefore\ t_1 = 6\ \mu s}$$

At $t_1 = 6\mu s$, Capacitor charged to 600 V and D_F starts Conducting.

(E)



$0 \leq t \leq 1\,\mu s$:-

$$E_{loss,1} = \int_0^{1\mu s} i_{CE}(t) \cdot v_{CE}(t)\, dt$$

$$= \int_0^{1\times10^{-6}} \left[60 - (40\times10^6)\,t\right]\left[40\times10^{12}\, t^2\right] dt.$$

By solving above integration

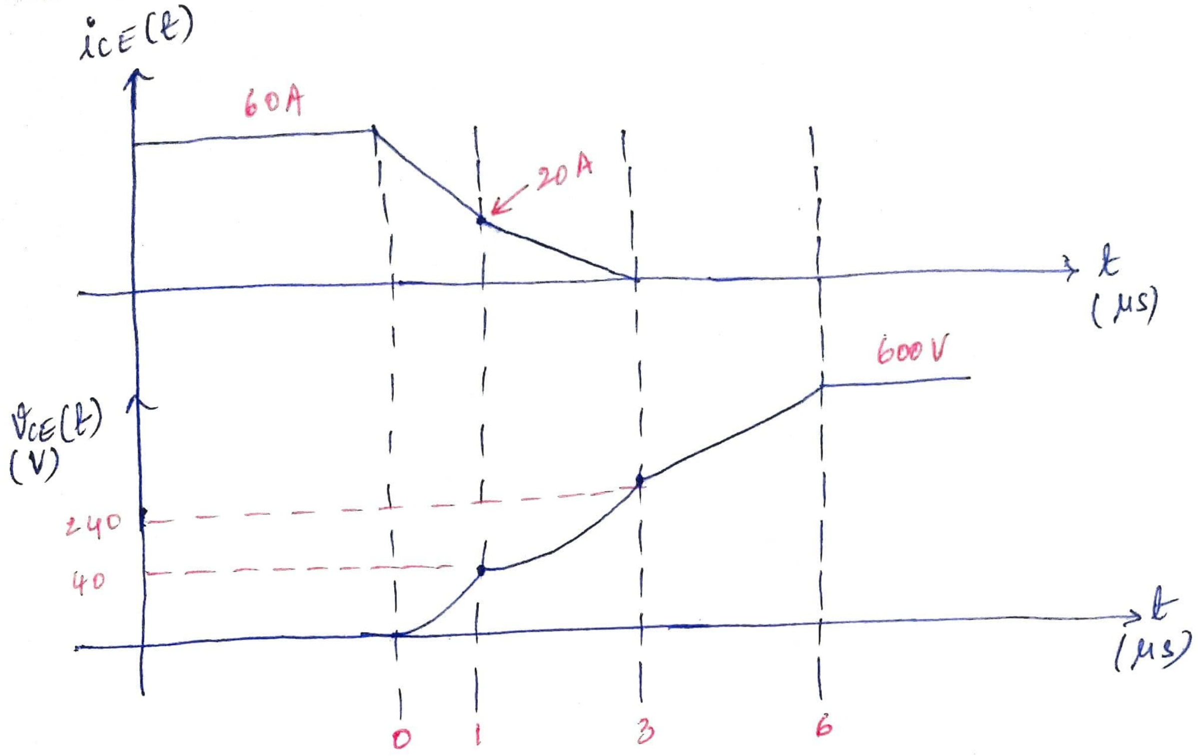
$$\boxed{E_{loss,1} = 0.4\, mJ}$$

$1\,\mu s \leq t \leq 3\,\mu s$:-

$$E_{loss,2} = \int_{1\mu}^{3\mu} i_{CE}(t) \cdot v_{CE}(t)\, dt$$

$$= \int_{1\mu}^{3\mu} \left[30 - (10\times10^6\, t)\right]\left[10\times10^{12}\, t^2 + (60\times10^6\, t) - 30\right] dt$$

(E)



<u>$0 \leq t \leq 1\,\mu s$:-</u>

$$E_{loss,1} = \int_{0}^{1\mu s} i_{CE}(t) \cdot v_{CE}(t)\, dt$$

$$= \int_{0}^{1\times10^{-6}} \left[60 - (40\times10^{6})\,t\right]\left[40\times10^{12}\,t^{2}\right] dt.$$

By solving above integration

$$\boxed{E_{loss,1} = 0.4\, mJ}$$

<u>$1\,\mu s \leq t \leq 3\,\mu s$:-</u>

$$E_{loss,2} = \int_{1\mu}^{3\mu} i_{CE}(t) \cdot v_{CE}(t)\, dt$$

$$= \int_{1\mu}^{3\mu} \left[30 - (10\times10^{6}\,t)\right]\left[10\times10^{12}\,t^{2} + (60\times10^{6}\,t) - 30\right] dt$$

By solving the above integration

$$E_{loss,2} = 2 mJ$$

$3\mu s \le t \le 6 \mu s$:- $E_{loss,3} = 0$

$$\therefore E_{loss} = E_{loss,1} + E_{loss,2} + E_{loss,3}$$

$$\boxed{E_{loss} = 2.4 mJ}$$

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(2) (A) $E_{OFF} = \int_0^{t_{OFF}} i_{sw}(t) \cdot V_{sw}(t) \, dt$

$0 \le t \le 200 ns$ $i_{sw}(t) = 50 - 0.25 \times 10^9 t$ A

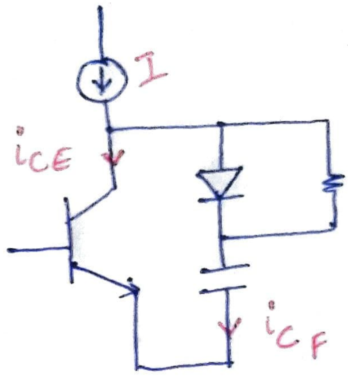
$$V_{sw}(t) = 200 V$$

$$\therefore E_{OFF} = \int_0^{200 \times 10^{-9}} (50 - 0.25 \times 10^9 t)(200) \, dt$$

By solving the above integration

$$\boxed{E_{OFF} = 1 mJ}$$

(B) $\underline{0 \le t \le 200\,ns}$:-



$$i_{CE}(t) = 50 - 0.25\times10^9 t \quad A$$

$$i_{CF}(t) = I - i_{CE}(t)$$

$$= 50 - 50 + 0.25\times10^9 t$$

$$i_{CF}(t) = 0.25\times10^9 t \quad A$$

$$\therefore \quad v_{CF}(t) = \frac{1}{C_F}\int_0^t i_{CF}(t)\,dt$$

$$= \frac{1}{C_F}\int_0^t 0.25\times10^9 t\,dt$$

$$v_{CF}(t) = \frac{0.25\times10^9}{2C_F}t^2 \quad V$$

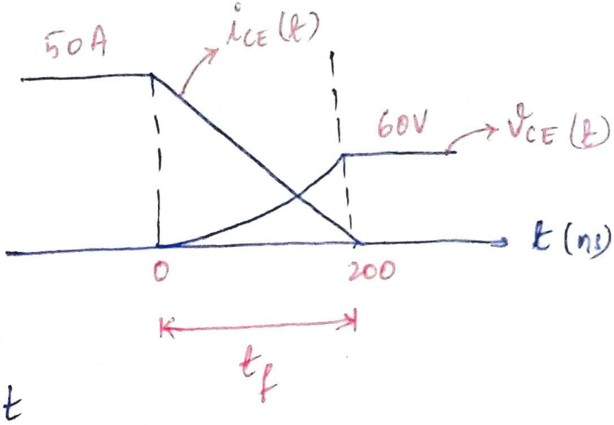
$$C_F = \frac{0.25\times10^9 t^2}{2 v_{CF}(t)}$$

At $t = 200\,ns$; $v_{CF}(t) = 60\,V$

$$\therefore \quad C_F = \frac{0.25\times10^9\times(200\times10^{-9})^2}{2\times60}$$

$$\boxed{C_F = 83.33\,nF}$$

(C)



$$E_{loss} = \int_0^{t_f} v_{CE}(t) \cdot i_{CE}(t)\, dt$$

$$\left.\begin{aligned} v_{CE}(t) &= v_{CF}(t) = \frac{0.25 \times 10^9}{2 C_F} t^2 \; V \\ i_{CE}(t) &= 50 - 0.25 \times 10^9 t \end{aligned}\right\} \text{From (B)}$$

$$\therefore E_{loss} = \int_0^{200 \times 10^{-9}} \frac{0.25 \times 10^9}{2 C_F} t^2 \left(50 - 0.25 \times 10^9 t\right) dt$$

By solving the above integration

$$\boxed{E_{loss} = 50 \mu J}$$

(D) $E_{stored} = \frac{1}{2} C_F V^2$

$$= \frac{1}{2} \times 83.33 \times 10^{-9} \times 200 \times 200$$

$$\boxed{E_{stored} = 1.67 mJ}$$